Engineering Mathematics - I · Unit_04 · Unit 4

# DRAFT — UE26MA141A · Unit_04 · Unit 4 · Gamma functions - definition, graph and properties

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

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M1

# Gamma functions - definition, graph and properties

## 1 Gamma functions - definition, graph and properties

Study focus: Gamma functions - definition, graph and properties. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

**Core explanation · 00:00–04:59**

I welcome all of you to special functions, which is your fifth unit in engineering mathematics one.

**Key details**

- In this unit, you are going to learn gamma function, beta function, and basis function.
- In this particular class, we are going to discuss definition of gamma function.
- Properties of gamma function and graph of gamma function.
- Gamma function is a definite integral whose integrant depends on one variable.
- It is also known as Euler's integral of second kind.
- It is the generalization of factorial notation from integes to real numbers.
- We can define gamma function as.
- Gamma of n is equal to integral zero to infinity t power minus x power n minus one dx n greater than c.
- Thus gamma n defines a function of n for positive values of n.
- From this result we can obtain the value of gamma.
- Then we get gamma 1 is equal to integral 0 to infinity e power minus x power 1 minus 1 dx.
- x power 1 minus 1 is x power 0 and we know x power 0 is 1, so it just reduces to integral 0 to infinity e power minus x bx.
- And we know that integration of e power minus x dx is equal to e power minus x divided by minus 1.
- Using the fact that e power minus infinity is equal to 0 and e power 0 is equal to 1, we can evaluate this to 1.
- Thus we have gamma of 1 is equal to 1.
- We can have an alternate form of gamma function.
- We can obtain it easily by using definition of gamma function.

- We know that gamma n is equal to integral 0 to infinity t power minus x power n minus 1 dx.
- If x is t square, dx is 2t d.
- Therefore, gamma n is equal to integral 0 to infinity e power minus t square t square to the power n minus 1 2 t dt.
- Observe here, after this substitution, the limits of integration do not change because when x is equal to 0, your t is also 0.
- When x is infinity, t is also infinite.
- This is another form of gamma function.
- While solving problems on gamma function, we can make use of any of these two forms according to our convenience.
- Next, we will discuss some properties of gamma function.
- First of them is reduction formula.
- It states that gamma of n plus 1 is equal to n gamma n where n is a real number.
- We can prove this easily by using the definition of gamma function, which is.
- Gamma n is equal to integral 0 to infinity e power minus x power n minus 1 dx.
- We want gamma of n plus 1.
- which is equal to integral 0 to infinity e power minus x power n dx.
- We have to evaluate this integral by using integration by parts.

**Example or application from the lecture**

- In this expression, let us substitute x is equal to t square.
- We know already that integration by parts is given by integral uv dx is equal to u into integral v dx minus integral of derivative of u or d by dx of u into integral of v dx.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

**Core explanation · 05:00–09:59**

When you take x power n is equal to u and e power minus x is equal to v, you will get it as x power n e power minus x divided by minus 1 plus n into integral 0 to infinity.

**Key details**

- E power minus x power n minus one dx.
- When you substitute the limits of integration for the first term,.
- E power minus infinity zero and e power zero is one.
- So first term becomes a zero and observe the second term along with n, you have integral zero to infinity, e power minus x, x power n minus one.
- dx which is nothing but equal to gamma n.
- When n is a negative non-integer, the formula used to find gamma of n is gamma n is equal to gamma of n plus 1 divided by n.
- Remember, gamma n is not defined for n is equal to 0.
- And it is also not defined when n is a negative integer.

- But we can find the value of gamma n if n is negative real number which is not a negative integer using this formula.
- When n is a positive integer, gamma of n plus one is equal to n factory.
- We can prove this easily by using reduction formula which states gamma of n plus 1 is equal to n gamma n.
- That is 1, because we know gamma 1 is equal to 1.
- When we put n is equal to 2, we get gamma of 3, that is 2 gamma 2, which is 2 factorial.
- When you put n is equal to 3, we get gamma 4, which is 3 gamma 3, that is 3 factorial.
- And when you put n is equal to 4, we get gamma 5, which is 4 gamma 4, that is 4 factor.
- Just look at the pattern here.
- Gamma two is one factorial, gamma three is two factorial, gamma four is three factorial, and gamma five is four factorial.
- Gamma of n plus 1 is equal to n factorial where n is a positive integer.
- Remember, gamma n is not defined when n is 0 or when n is a negative integer.
- Gamma half value we can find now.
- Gamma half is equal to square root of pi.
- We can prove this by considering alternate form of gamma function, which is gamma n is equal to 2 times integral 0 to infinity e power minus x square x to the power 2n minus 1 dx.
- Which can also be written as two times integral 0 to infinity e power minus y square dy.
- Because since we are dealing with definite integrals, here x is a dummy variable.
- You can write it as x or y or t or any variable since we are dealing with definite integrals.
- which is equal to four times integral 0 to infinity 0 to infinity e power minus x square plus y square dx d y.
- Then x square plus y square is equal to r square.

**Example or application from the lecture**

- Let us put n is equal to 1 in this formula, we get gamma 2 is equal to 1, gamma 1.
- Let us put n is equal to half in this expression.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

**Core explanation · 10:00–14:59**

And dx dy is equal to r dr d theta.

**Key details**

- And observe the limits for x and y.
- Both are varying from 0 to infinity.
- That means both x and y are positive.
- So our region of integration is the first quadrant.
- In this first quadrant, r value varies from 0 to infinity and theta value varies from 0 to pi by 2.
- Then it can be written as four into integral 0 to pi by 2 b theta into integral 0 to infinity e power minus r square r br.

- When you integrate the first integral 0 to pi by 2 d theta, integration of d theta is theta, substitute the limits of integration, it reduces to 4 into pi by 2.
- into integral 0 to infinity e power minus r square r dr.
- That is equal to 2 by into minus 1 by 2 e power minus r square limits vary from 0 to infinity.
- Thus, we get gamma half the whole square is equal to pi, so we can conclude gamma half is equal to square root of pi, which is a very important result.
- This is the graph of gamma function.
- Observe here, gamma function is continuous for positive values of n and it is not continuous for negative values of n.
- We can find or you can plot this curve by using the properties gamma of n plus 1 is equal to n gamma n.
- Gamma n is equal to gamma n plus 1 divided by n.
- This is for negative real numbers which are not indigenous.
- And gamma half is equal to square root of pi.
- With these properties and the table, try to get the shape.
- This is very interesting thing.
- You can really do it.
- You will enjoy doing it.
- With this, we will end the session.

**Example or application from the lecture**

- Before that, just have a glance at this table where gamma n values are given for various values of n between 1 and 2.
- And also referring to the table which is given in the previous slide.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Gamma functions - definition, graph and properties transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Gamma functions - definition, graph and properties transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Gamma functions - definition, graph and properties transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps. |
| Gamma functions - definition, graph and properties transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_04/AV_Summaries/001_Gamma_functions_-_definition_graph_and_properties__c183e86bfc3ac0a2.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

### Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.

3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00002 | 3 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare... |
| SRC00003 | 3 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
|---|---|---|
| SRC00004 | 3 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare... |
| SRC00005 | 3 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare...<br>t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare... |

# C Revision and retrieval practice